



SQL SUBLANGUAGES AND FUNCTIONS



OVERVIEW

- The Sublanguages of SQL
 - Data Definition Language
 - Data Manipulation Language
 - Data Query Language
 - Data Control Language
 - Transaction Control Language
- Scalar Functions
- Aggregate Functions

WHAT ARE SQL SUBLANGUAGES?

- SQL has many commands and keywords; to keep them all straight, they are split into sublanguages
- Here are the 5 you should know
 - Data Definition Language (DDL) - Create, Alter, Drop
 - Data Manipulation Language (DML) - Update, Delete, Insert
 - Data Query Language (DQL) - Select
 - Data Control Language (DCL) - Grant, Revoke
 - Transaction Control Language (TCL) - Commit, Rollback

DATA DEFINITION LANGUAGE

- Data Definition Language is the language for defining the database schema.
- DDL Commands:
 - Create – used to create database objects such as tables
 - Drop – used to delete objects from the database
 - Alter – used to alter the structure of the database
 - Truncate – used to remove all records from a table, all spaces allocated for the records are removed as well
 - Comment – used to add comments to the data dictionary
 - Rename – used to rename an object existing in the database

DATA MANIPULATION LANGUAGE

- Data Manipulation Language is used for manipulating the data in the database.
- DML Commands:
 - Insert – used to insert data into a table
 - Update – used to update existing data within a table
 - Delete – used to delete records(rows) from a table
 - Lock – table control concurrency

DATA QUERY LANGUAGE

- Data Query Language is used for retrieving data from your database through queries
- DQL Command:
 - Select – used to start all queries for retrieving data from the database

DATA CONTROL LANGUAGE

- Data Control Language is used for defining access rights and permission controls of the database
- DCL Commands:
 - Grant – This gives users access privileges to the database
 - Revoke – This withdraws the user's access privileges previously given by the GRANT command

TRANSACTION CONTROL LANGUAGE

- Transaction Control Language is used for managing database transactions
- TCL Commands:
 - Commit – commits a transaction
 - Rollback – rolls back a transaction in case any error occurs
 - Savepoint – Sets a save point within a transaction

SCALAR FUNCTIONS

- Scalar functions take a set of number inputs and give one expected output
- Examples:
 - Ceil() rounds a number up to the next whole number
 - Concat() combines two strings
 - Substr() returns a substring
 - Length() returns the length of a string

SCALAR FUNCTION EXAMPLE

- The following example adds the string 'Dept:' to the DepartmentName field
- `SELECT FirstName, LastName, CONCAT ('Dept:' , DepartmentName) FROM employees`

AGGREGATE FUNCTIONS

- Aggregate functions combine rows of data into a single cell
- Examples:
 - Sum() sums up the numeric inputs
 - Avg() averages the inputs
 - Count() increments once per input
 - Max() contains the value of the largest input

AGGREGATE FUNCTION EXAMPLE

- The following example will calculate the average salary of the employees and group them based on the department name
- `SELECT DepartmentName,AVG (Salary) FROM Employees GROUP BY DepartmentName`